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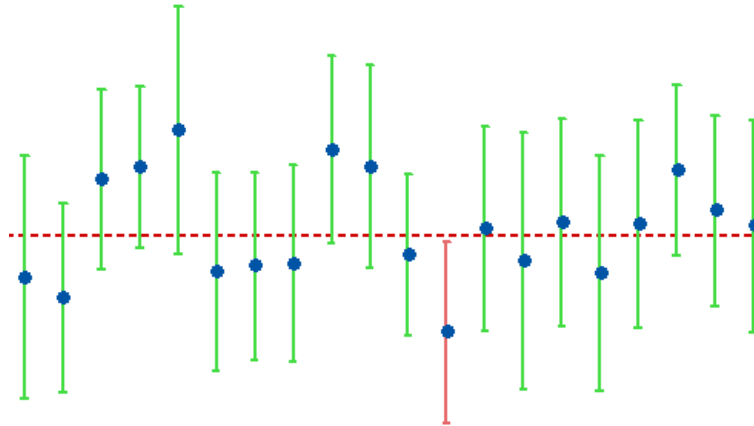
Introduction to Data Science

Lecture 12: Confidence Interval (continued) + Midterm Review

Outline

- Confidence Intervals (*continued*)
 - Limitation of point estimation
 - Interval Estimation
 - Central Limit Theorem
 - CI for the mean
- Midterm Review

Confidence Intervals (*continued*)



Limitation of Point Estimation

- Point estimation provides a **single best guess** for an unknown population parameter
 - Example: Estimating the population mean using the sample mean
- A point estimate does *NOT* capture *uncertainty* or variability in the estimate

Example of Point Estimation

- MLE is a widely used approach for point estimation, which finds the parameter value that makes the observed data **most likely**
- Given a sample $X_1, X_2, \dots, X_n$, we choose a parameter θ to **maximize the likelihood function**

$$L(\theta) = P(X_1, X_2, \dots, X_n \mid \theta)$$

Example of MLE: Estimating Mean of a Normal

- Suppose the sample $X_1, X_2, \dots, X_n$ is iid from $N(\mu, \sigma^2)$
- To estimate μ , we write the likelihood function

$$L(\mu) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(X_i - \mu)^2}{2\sigma^2}\right)$$

- It turns out the **sample mean** maximizes the likelihood function

$$\hat{\mu}_{MLE} = \frac{1}{n} \sum_{i=1}^n X_i$$

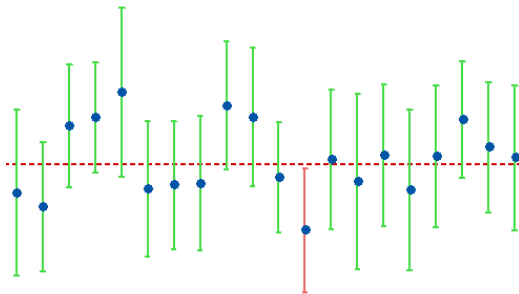
- However, it is unclear how accurate this estimator is
 - This is where **interval estimation** helps!

Point vs. Interval Estimation

- A single point estimate is easier to obtain, but does not quantify how much uncertainty exists in the estimation
- **Interval estimation** provides an interval that contains plausible values for a parameter
 - **E.g.:** A 95% confidence interval (70, 80) means we are **95% confident** that the true population mean lies between 70 and 80.
 - Advantage: Captures uncertainty and reflects variability in the data, making it more informative than point estimation.

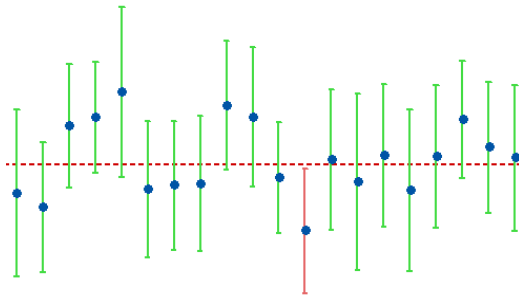
Confidence Interval: Definition

- A **confidence interval (CI)** is a range of values of the form $[a,b]$, derived from sample data, that is likely to contain the true population with a specified confidence
 - $[a, b]$ is a *random* interval, since it depends on the sample data



Confidence Interval: Interpretation

- A *95% confidence interval* for a population mean indicates that
 - If we were to repeat the sampling process many times and compute a confidence interval each time, then approximately 95% of those intervals would contain the true population mean



- But usually, we do not repeat the above process, we interpret the 95% as
 - $[a, b]$ contains the true value with *confidence* 95%

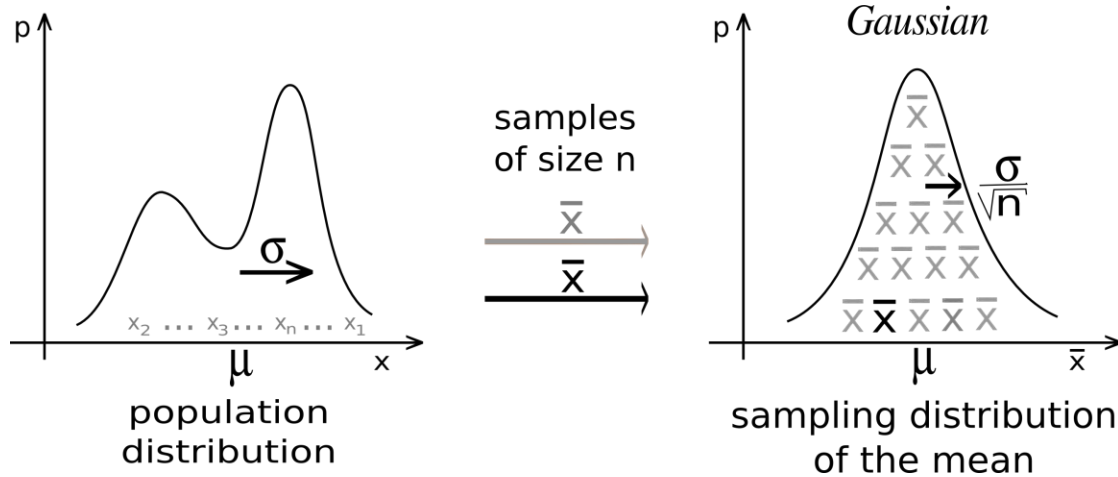
How to Estimate a Confidence Interval?

- Thanks to the **Central Limit Theorem (CLT)**, we can apply normal distribution properties to construct confidence intervals
- *[CLT] No matter what the population distribution is, for **large n** , the sample mean follows a **normal distribution***

$$\frac{\sqrt{n}(\bar{X} - \mu)}{\sigma} \sim N(0, 1)$$

- Here:
 - $\bar{X} = \frac{X_1 + X_2 + \dots + X_n}{n}$ is the sample mean
 - μ is the population mean, and σ is the population standard deviation

Central Limit Theorem



- No matter what the true distribution is, the distribution of the **sample mean** $\bar{X}$ will be very close to a **normal distribution**, as long as the sample size is **large** (*rule of thumb: 30-sample size for a one-dimensional variable*)

Confidence Interval for the Population Mean with Known Variance

- Suppose we would like to estimate the mean of a random variable X with variance σ^2
- We are given a sample of size n : $X_1, X_2, \dots, X_n$
- *Question: Estimate the mean of the random variable*

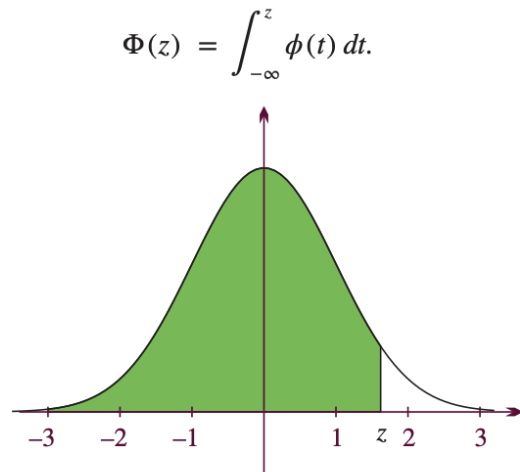
Confidence Interval for the Mean: Derivations

- From the CLT, $\frac{\sqrt{n}(\bar{X}-\mu)}{\sigma} \sim N(0,1)$
- Hence, if we let $\Phi(x)$ be the CDF of a standard normal distribution, then

$$P(-a \leq \frac{\sqrt{n}(\bar{X}-\mu)}{\sigma} \leq b) \approx \Phi(b) - \Phi(-a)$$

- The event above is equivalent to

$$\bar{X} - \frac{b \sigma}{\sqrt{n}} \leq \mu \leq \bar{X} + \frac{a \sigma}{\sqrt{n}}$$



Confidence Interval for the Mean: Derivations

- It follows that when n is large,

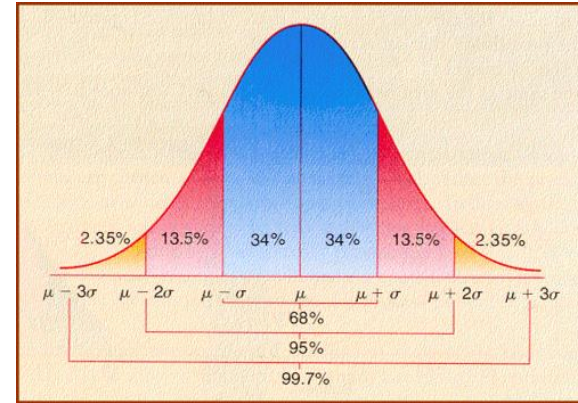
$$P\left(\bar{X} - \frac{b\sigma}{\sqrt{n}} \leq \mu \leq \bar{X} + \frac{a\sigma}{\sqrt{n}}\right) \approx \Phi(b) - \Phi(-a)$$

- By symmetry, $\Phi(-a) = 1 - \Phi(a)$

- Set $a = b$, the interval

$$\left[\bar{X} - \frac{a\sigma}{\sqrt{n}}, \bar{X} + \frac{a\sigma}{\sqrt{n}}\right]$$

contains μ with probability approximately $2\Phi(a) - 1$



Confidence Interval for the Mean: Derivations

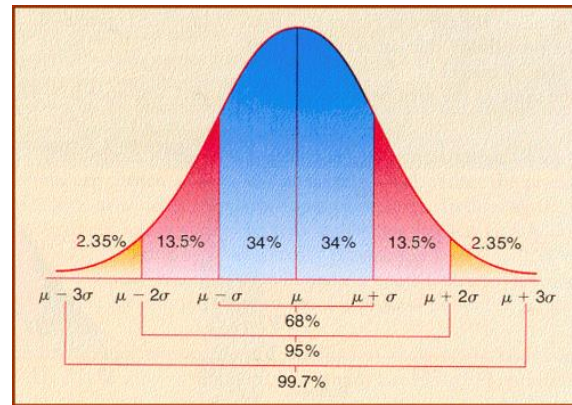
- Let $Z \sim N(0,1)$. Define $z_{\alpha/2}$ such that

$$P(-z_{\alpha/2} \leq Z \leq z_{\alpha/2}) = 1 - \alpha,$$

- E.g., when $\alpha = 0.05$, $z_{0.025} = 1.96$
- Then by setting $a = z_{\alpha/2}$, the interval

$$\left[\bar{X} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}}, \bar{X} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \right]$$

is a $100(1-\alpha)\%$ CI on the mean μ



Confidence Interval for the Mean

A 100(1- α)% Confidence Interval (CI) is given by

$$[\bar{X} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}}, \bar{X} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}}]$$

- The length of CI is affected by several factors:
 - As the sample size **n increases**, the length of CI **decreases**
 - As the variance **σ^2 increases**, the length of CI **increases**
 - As the confidence level (1- α) increases (or **α decreases**), the length of CI **increases**
- CI is random, as $\bar{X}$ is random

Example: Blind Box

- A blind-box seller tells you: with a probability p , you will get a Harry-Potter
- You bought 100 boxes and got 49 Harry-Potters
- What's your point estimate for p ?

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 - $\hat{p} = 0.49$

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- What's your *CI estimate* for p ?
 - The standard deviation of a Bernoulli variable is $\sigma = \sqrt{p(1 - p)}$
 - Using $\hat{p}$, we estimate σ as $\sqrt{0.49(1 - 0.49)}$
 - Plug into the formula using $z_{0.05/2} \approx 1.96$:

$$\begin{aligned} & \left[0.49 - 1.96 \frac{\sqrt{0.49(1-0.49)}}{\sqrt{100}}, \quad 0.49 + 1.96 \frac{\sqrt{0.49(1-0.49)}}{\sqrt{100}} \right] \\ & \approx [0.49 - 0.098, 0.49 + 0.098] \end{aligned}$$

Exercise: Physical Measurements

- A manufacturer produces piston rings for an automobile engine. It is known that ring diameter is normally distributed with $\sigma = 0.001$ millimeters.
- A random sample of 15 rings has a sample mean diameter of 74.036mm.
- Construct a 99% confidence interval estimate for the mean piston ring diameter. ($z_{0.005} = 2.58, z_{0.01} = 2.33$)

Exercise: Physical Measurements

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✓ Using $74.036 \pm 2.58 \frac{0.001}{\sqrt{15}}$ yields a 99% CI [74.0353, 74.0367]

✓ Using $74.036 \pm 2.33 \frac{0.001}{\sqrt{15}}$ yields a 98% CI [74.0354, 74.0366]

Value of Interval Estimation

- A blind-box seller tells you: with a probability higher than **60%**, you will get a Harry-Potter
- You bought 100 boxes and got 49 Harry-Potter
- Did the seller lie to you?

Value of Interval Estimation

- A blind-box seller tells you: with a probability higher than 60%, you will get a Harry-Potter
- You bought 100 boxes and got 49 Harry-Potter
- Did the seller lie to you?

➤ Point estimation: 49%.

➤ Interval estimation: the 95% CI is $[0.392, 0.588]$

✓ With 95% confidence, the CI does not contain 60%. So YES!

✓ You are 95% confident that the seller lied to you

Hypothesis Test!

Discussion: The best interval?

- With probability $\Phi(b) - \Phi(-a)$, μ is within $[\bar{X} - \frac{b\sigma}{\sqrt{n}}, \bar{X} + \frac{a\sigma}{\sqrt{n}}]$
- Fix $\Phi(b) - \Phi(-a)$, there are numerous choices of **a** and **b**
- If the population mean μ has an **upper bound**, say U
 - when $\bar{X}$ is too close to U, choose **a** such that $\bar{X} + \frac{a\sigma}{\sqrt{n}} = U$

This will not be included in the exam

Discussion: The best interval?

- With probability $\Phi(b) - \Phi(-a)$, μ is within $[\bar{X} - \frac{b\sigma}{\sqrt{n}}, \bar{X} + \frac{a\sigma}{\sqrt{n}}]$
- Fix $\Phi(b) - \Phi(-a)$, there are numerous choices of **a** and **b**
- If the population mean μ has an **upper bound**, say U
 - when $\bar{X}$ is too close to U, choose **a** such that $\bar{X} + \frac{a\sigma}{\sqrt{n}} = U$
- If the population mean μ has a **lower bound**, say L
 - when $\bar{X}$ is too close to L, choose **b** such that $\bar{X} - \frac{b\sigma}{\sqrt{n}} = L$

This will not be included in the exam